

ToneCanvas: Visually Painting Character Tone for Narrative Consistency

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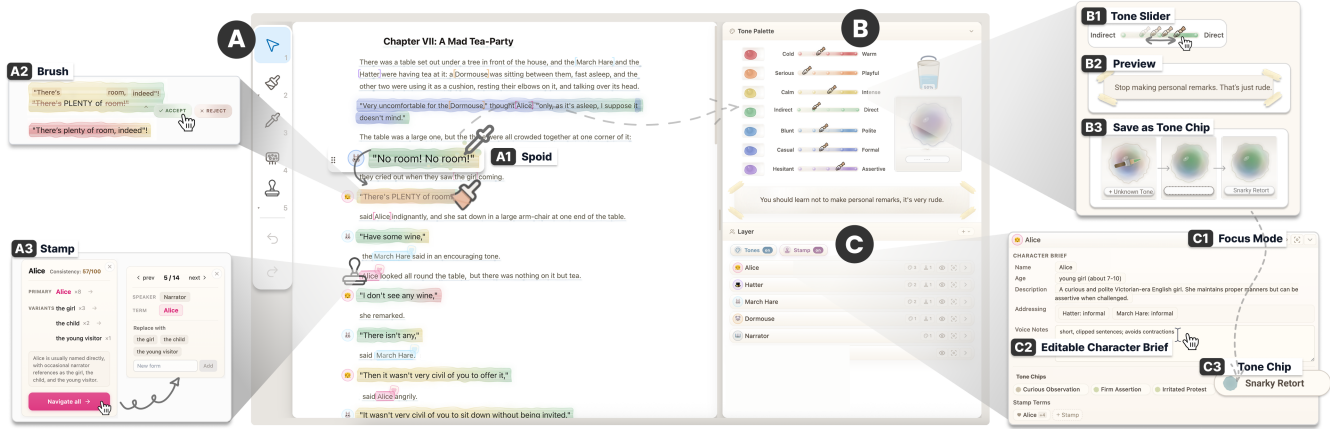


Figure 1: ToneCanvas supports character-tone revision through comparison, specification, and reference. (A) Persistent watercolor overlays make character tone visible on the manuscript, with painting-inspired tools for tone sampling, revision, and terminology review. (B) The Tone Palette adjusts tone through independent dimensions with a live preview. (C) Character Layers gather dispersed dialogue, character references, and reusable Tone Chips for cross-passage revision.

Abstract

Revising long-form fiction requires writers to maintain each character's voice across dialogue that is often dispersed throughout a manuscript. Detecting gradual changes in how a character speaks therefore depends heavily on rereading, memory, and external notes. We present ToneCanvas, an interactive writing interface that uses persistent watercolor gradients to support character-centered tone comparison and revision within and across passages. A formative study with five professionals involved in fiction revision and translation identified four recurring workflow breakdowns, which informed the design of character-focused dialogue views, adjustable tone dimensions, and integrated character references. This work contributes an empirical account of character-tone revision, corresponding design opportunities, and an interactive prototype that instantiates them.

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Narrative Consistency; Character Voice; Visual Text Analytics; LLM-Augmented Interface

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1 Introduction and Background

Revising long-form fiction requires writers to maintain each character's voice across dialogue that is often scattered throughout the manuscript. For example, a writer may decide that a villain should gradually become warmer after a narrative turning point, yet identifying every dialogue passage that should reflect this change requires substantial rereading, cross-referencing with external character notes, and relying on memory across the manuscript [4, 5].

We focus on tone as a central aspect of character voice: how a character speaks through register, warmth, directness, politeness, intensity, rhythm, and interpersonal stance [1–3, 8, 13, 15]. Whereas emotion may fluctuate between scenes, tone contributes to a recognizable speaking style while evolving over a narrative. Revising

tone is therefore inherently comparative: writers must judge each passage against the same character’s dialogue elsewhere in the manuscript.

Existing writing tools provide limited support for reviewing character tone across a manuscript. LLM-based co-writing interfaces support drafting, rewriting, character development, and structured control over generated text, but generally begin with a passage already selected by the writer [9, 10, 14, 16, 19, 21]. Visual writing and narrative analysis systems externalize feedback, entities, timelines, narrative trajectories, character emotions, and relationships, but focus primarily on analyzing narrative content rather than revising character voice across passages [6, 7, 12, 17, 18, 20]. Textshop takes a more direct approach by representing tone as color and making it editable through dimension controls and brush-based interactions, yet it likewise operates primarily on selected text [11]. ToneCanvas combines these directions by extending tone-as-color from local text manipulation to persistent, per-character comparison across a manuscript.

Based on our formative study, we developed ToneCanvas to support the discovery and correction of character-voice inconsistencies. This work contributes (1) an empirical account of character-tone revision in long-form fiction, (2) four design opportunities derived from recurring workflow breakdowns, and (3) an interactive prototype that realizes these opportunities through persistent tone visualization, parameterized control, and integrated character references.

2 Formative Study

Building on prior formative studies of writers’ needs for character analysis and visualization [6, 7, 11], we shifted the focus from analyzing completed narratives to discovering and correcting voice drift during revision. We recruited five writing professionals through writing communities and subsequent snowball sampling. Participants had 5–15 years of experience in fiction revision or translation and regularly worked with extended, multi-character narratives. During 60-minute semi-structured interviews, participants walked us through their revision workflows, recalled difficult revision episodes, and critiqued early interface sketches. Our analysis identified four recurring breakdowns in character-centered revision.

Tone is distributed and difficult to review. Writers relied on rereading and memory to compare how a character spoke across dispersed dialogue. F3, who writes long-running martial-arts web novels, said: “I can’t re-read everything. But there’s no other way.” Gradual changes were especially difficult to detect. F5 recalled a villain whose speech “gradually softened over twenty chapters” before a reader noticed.

Writers detect inconsistency but struggle to articulate it. Even after locating a problematic passage, writers often struggled to identify which aspect of the character’s tone had changed. F4 captured the gap between intuition and action: “I can tell something’s off, but then I’m stuck thinking—what should I change it to?” Prompts such as “make this sound colder” often shifted formality, warmth, and directness together, even when only one was intended.

Character voice references remain outside the manuscript. The writers maintained character sheets, glossaries, and reference

passages that describe how each character should speak. F2 recorded “vocabulary level, sentence length, how often they show emotion, and signature endings” for major characters. Consulting these references required repeatedly switching between the manuscript and supporting documents; participants described this switching as disruptive to revision itself, as each lookup forced them to re-read the passage to re-establish where they were and what they had intended to change.

Character terminology varies across long drafts. Writers also struggled to maintain consistent references to characters and locations. A character might appear under a full name, shortened form, title, alias, or romanization variant. F3 kept a spreadsheet to track these references, but noted “going back and forth is painful,” pointing to a need to review variants directly within the manuscript. Such referential variation is itself part of voice: shifts among a full name, nickname, title, or honorific signal changes in register and interpersonal stance [2], so uncontrolled variation can both obscure the tonal signal writers are tracking and read as unintended tonal drift.

These findings suggest four design opportunities: surface distributed tone patterns, decompose tone into adjustable dimensions, integrate character references, and review terminology variants in context.

3 ToneCanvas Prototype

Character-tone revision requires comparing how a character speaks across dispersed dialogue throughout a manuscript. ToneCanvas supports this process with persistent watercolor overlays on the text. Motivated by the formative study’s call for in-context visualization, we chose translucent watercolor because it remains visible without obscuring the manuscript and can blend multiple tonal qualities. Each dialogue block is characterized along seven fixed bipolar axes drawn from prior work and the interviews: warmth, playfulness, intensity, directness, politeness, formality, and assertiveness. The two most prominent appear as a left-to-right gradient, supporting cross-passage comparison before revision.

Compare character tone across passages. ToneCanvas renders dialogue blocks with persistent watercolor overlays that encode prominent tonal dimensions. Color differences draw attention to passages whose tone departs from surrounding dialogue, while **Focus Mode** gathers one character’s dialogue into a single view for inspecting gradual shifts and recurring voice patterns.

Specify and revise tone. Following structured LLM interfaces that expose an editable design space [16, 21], the Tone Palette lets writers inspect and adjust tone through independently controllable dimensions—separating aspects such as formality, warmth, and directness that writers in our formative study reported shifting together unintentionally—while previewing changes on a character-specific reference sentence. **Spoid** samples a passage’s tonal profile into the palette; **Brush** applies a target tone with an inline diff for writer approval; **Sponge** explores smoother transitions between adjacent passages; and **Tone Chips** save reusable tone presets for each character.

Reference character information during revision. Each Character Layer stores a Character Brief containing voice notes, addressing style, and other character-specific references beside the

manuscript. These references are automatically incorporated into rewriting so writers do not need to repeatedly consult external documents.  **Stamp** groups names, aliases, titles, and recurring terms, allowing writers to review and resolve terminology variants directly within the manuscript.

4 Conclusion and Future Work

ToneCanvas makes character tone inspectable across long manuscripts by connecting visual comparison with direct revision. Grounded in a formative study with experienced fiction writers and previous literature, the prototype supports comparing character tone, specifying tonal changes, and referencing character information during revision. Future work will evaluate ToneCanvas in writers' own drafts to understand how persistent visual comparison supports intentional character development over extended writing sessions. We also plan to explore macro-level overviews that summarize tonal trajectories per scene or act, and addressee-dependent tone, as a character's voice may legitimately shift with the interlocutor.

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References

- [1] Douglas Biber. 2006. University language: A corpus-based study of spoken and written registers.
- [2] Penelope Brown. 1987. *Politeness: Some universals in language usage*. Vol. 4. Cambridge university press.
- [3] Cristian Danescu-Niculescu-Mizil, Moritz Sudhof, Dan Jurafsky, Jure Leskovec, and Christopher Potts. 2013. A computational approach to politeness with application to social factors. In *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Hinrich Schuetze, Pascale Fung, and Massimo Poesio (Eds.). Association for Computational Linguistics, Sofia, Bulgaria, 250–259. <https://aclanthology.org/P13-1025/>
- [4] David K Farkas. 1985. The concept of consistency in writing and editing. *Journal of Technical Writing and Communication* 15, 4 (1985), 353–364. doi:10.2190/T6EM-UTT0-EL6J-59N9
- [5] Linda Flower, John R Hayes, Linda Carey, Karen Schriver, and James Stratman. 1986. Detection, diagnosis, and the strategies of revision. *College Composition & Communication* 37, 1, 16–55. doi:10.58680/coc198611245
- [6] Md Naimul Hoque, Bhavya Ghai, and Niklas Elmqvist. 2022. DramatVis Personae: Visual Text Analytics for Identifying Social Biases in Creative Writing. In *Proceedings of the 2022 ACM Designing Interactive Systems Conference (Virtual Event, Australia) (DIS '22)*. Association for Computing Machinery, New York, NY, USA, 1260–1276. doi:10.1145/3532106.3533526
- [7] Md Naimul Hoque, Bhavya Ghai, Kari Kraus, and Niklas Elmqvist. 2023. Portrayal: Leveraging NLP and Visualization for Analyzing Fictional Characters. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference (Pittsburgh, PA, USA) (DIS '23)*. Association for Computing Machinery, New York, NY, USA, 74–94. doi:10.1145/3563657.3596000
- [8] Ken Hyland. 2005. Stance and engagement: A model of interaction in academic discourse. *Discourse studies* 7, 2 (2005), 173–192. doi:10.1177/1461445605050365
- [9] Mina Lee, Katy Ilonka Gero, John Joon Young Chung, Simon Buckingham Shum, Vipul Raheja, Hua Shen, Subhashini Venugopalan, Thimo Wambsganss, David Zhou, Emad A. Alghamdi, Tal August, Avinash Bhat, Madiha Zahrah Choksi, Senjuti Dutta, Jin L.C. Guo, Md Naimul Hoque, Yewon Kim, Simon Knight, Seyed Parsa Neshaei, Antonette Shibani, Disha Shrivastava, Lila Shroff, Agnia Sergeyuk, Jessi Stark, Sarah Sterman, Sitong Wang, Antoine Bosselut, Daniel Buschek, Joseph Chee Chang, Sherol Chen, Max Kreminski, Joonsuk Park, Roy Pea, Eugenia Ha Rim Rho, Zejiang Shen, and Pao Siangliulue. 2024. A Design Space for Intelligent and Interactive Writing Assistants. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 1054, 35 pages. doi:10.1145/3613904.3642697
- [10] Mina Lee, Percy Liang, and Qian Yang. 2022. CoAuthor: Designing a Human-AI Collaborative Writing Dataset for Exploring Language Model Capabilities. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 388, 19 pages. doi:10.1145/3491102.3502030
- [11] Damien Masson, Young-Ho Kim, and Fanny Chevalier. 2025. Textshop: Interactions Inspired by Drawing Software to Facilitate Text Editing. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). Association for Computing Machinery, New York, NY, USA, Article 1087, 14 pages. doi:10.1145/3706598.3713862
- [12] Damien Masson, Zixin Zhao, and Fanny Chevalier. 2025. Visual Story-Writing: Writing by Manipulating Visual Representations of Stories. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25)*. Association for Computing Machinery, New York, NY, USA, Article 70, 15 pages. doi:10.1145/3746059.3747758
- [13] Saif Mohammad. 2018. Obtaining Reliable Human Ratings of Valence, Arousal, and Dominance for 20,000 English Words. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Iryna Gurevych and Yusuke Miyao (Eds.). Association for Computational Linguistics, Melbourne, Australia, 174–184. doi:10.18653/v1/P18-1017
- [14] Hua Xuan Qin, Shan Jin, Ze Gao, Mingming Fan, and Pan Hui. 2024. CharacterMeet: Supporting Creative Writers' Entire Story Character Construction Processes Through Conversation with LLM-Powered Chatbot Avatars. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 1051, 19 pages. doi:10.1145/3613904.3642105
- [15] Sudha Rao and Joel Tetreault. 2018. Dear Sir or Madam, May I Introduce the GYAFC Dataset: Corpus, Benchmarks and Metrics for Formality Style Transfer. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, Marilyn Walker, Heng Ji, and Amanda Stent (Eds.). Association for Computational Linguistics, New Orleans, Louisiana, 129–140. doi:10.18653/v1/N18-1012
- [16] Sangho Suh, Meng Chen, Bryan Min, Toby Jia-Jun Li, and Haijun Xia. 2024. Luminate: Structured Generation and Exploration of Design Space with Large Language Models for Human-AI Co-Creation. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 644, 26 pages. doi:10.1145/3613904.3642400
- [17] Katie Watson, Samuel S. Sohn, Sasha Schriber, Markus Gross, Carlos Manuel Muniz, and Mubbasir Kapadia. 2019. StoryPrint: an interactive visualization of stories. In *Proceedings of the 24th International Conference on Intelligent User Interfaces (Marina del Ray, California) (IUI '19)*. Association for Computing Machinery, New York, NY, USA, 303–311. doi:10.1145/3301275.3302302
- [18] Catherine Yeh, Tara Menon, Robin Singh Arya, Helen He, Moira Weigel, Fernanda Viégas, and Martin Wattenberg. 2025. Story Ribbons: Reimagining Storyline Visualizations with Large Language Models. *IEEE Transactions on Visualization and Computer Graphics* (2025). doi:10.1109/TVCG.2025.3634265
- [19] Ann Yuan, Andy Coenen, Emily Reif, and Daphne Ippolito. 2022. Wordcraft: Story Writing With Large Language Models. In *Proceedings of the 27th International Conference on Intelligent User Interfaces (Helsinki, Finland) (IUI '22)*. Association for Computing Machinery, New York, NY, USA, 841–852. doi:10.1145/3490099.3511105
- [20] Chao Zhang, Kexin Ju, Zhuolun Han, Yu-Chun Grace Yen, and Jeffrey M. Rzeszotarski. 2025. Synthia: Visually Interpreting and Synthesizing Feedback for Writing Revision. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25)*. Association for Computing Machinery, New York, NY, USA, Article 88, 16 pages. doi:10.1145/3746059.3747703
- [21] Zheng Zhang, Jie Gao, Ranjodh Singh Dhaliwal, and Toby Jia-Jun Li. 2023. VISAR: A Human-AI Argumentative Writing Assistant with Visual Programming and Rapid Draft Prototyping. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (San Francisco, CA, USA) (UIST '23)*. Association for Computing Machinery, New York, NY, USA, Article 5, 30 pages. doi:10.1145/3586183.3606800